and out of the parenchymatous cells of the pulvinus (Figure 15.18).

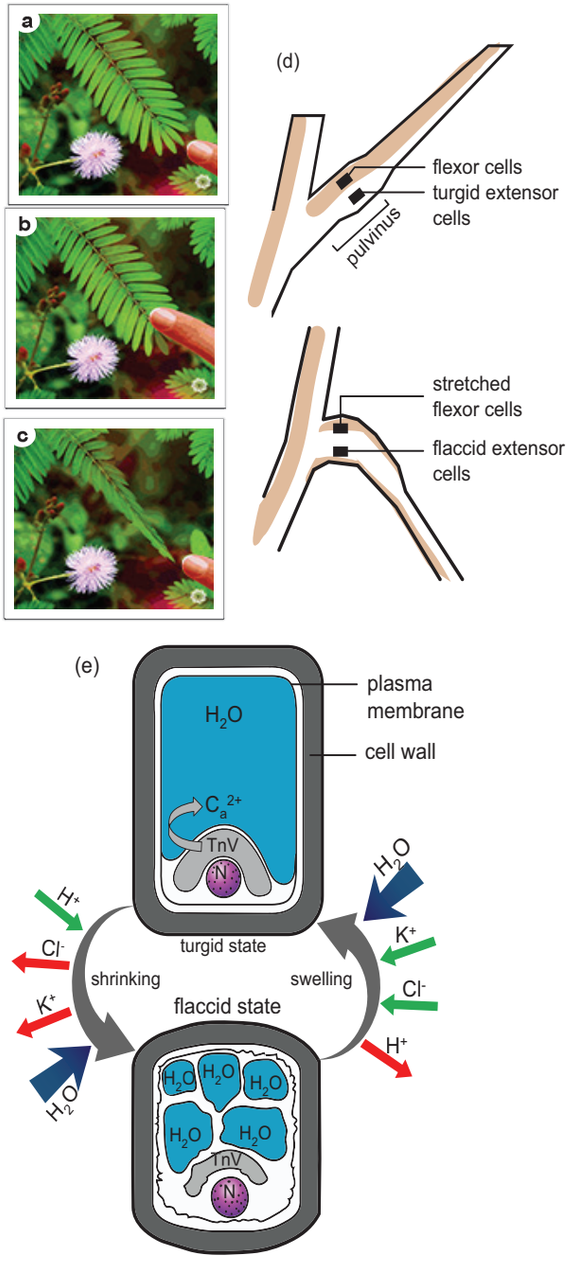


Figure 15.18: Mechanism of Seismonastic movement in *Mimosa pudica*

3. Thigmonastic movement

The movements found in the leaves of *Drosera* and *Dionaea* (Venus fly trap) result in response to the touch stimulus of insects. As soon as an insect sits on the leaf the cilia curve inward to trap the insect and trigger hairs are activated. Similarly, in dionaea, the two halves of the leaf curve upwards along the midrib. These parts of the leaves come to their normal position after the insect has been digested (Figure 15.19).



Figure 15.19: Thigmonasty in *Dionaea*

II Physical Movement (Hygroscopic Movements)

Physical movements are those which are found in dead parts of the plants and they are not related to any irritability of the protoplasm. They are also called **hygroscopic movements** or **mechanical movements**. Dispersal of spores and seeds, dehiscence of sporangia, bursting of seeds and movement of elaters are the examples of physical or hygroscopic movement.

15.4 Photoperiodism

Trees take several years for initiation of flowering whereas an annual herb flowers within few months. Each plant requires a specific time period to complete their vegetative phase which will be followed by reproductive phase as per their internal control points through Biological Clock. The physiological mechanisms in relation to flowering are controlled by (i) light period (Photoperiodism) and (ii) temperature (Vernalization). The physiological change on flowering due to relative length of light and darkness (photoperiod) is called **Photoperiodism**. The term photoperiodism was coined by **Garner** and **Allard** (1920) when they observed this in 'Biloxi' variety of soybean (*Glycine max*) and 'Maryland mammoth' variety of tobacco (*Nicotiana tabacum*). The photoperiod required to induce flowering is called **critical day length**. Maryland mammoth (tobacco variety) requires 12 hours of light and cocklebur (*Xanthium pensylvanicum*) requires 15.05 hours of light for flowering.

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